

ORBIT WORKFORCE MANAGEMENT SYSTEM APP

BIT 4107 – MOBILE APPLICATION DEVELOPMENT

PROJECT DOCUMENTATION

CHAPTER ONE: INTRODUCTION

Hardware Requirements

- Computer/Laptop
 - Android Smartphone
 - Internet Connection
-

Software Requirements

- Android Studio
 - Flutter SDK
 - Dart SDK
 - SQLite Database
 - Supabase
 - GitHub
 - Windows Operating System
-

System Framework

The Orbit Workforce Management System follows a three-layer architecture.

USER LAYER

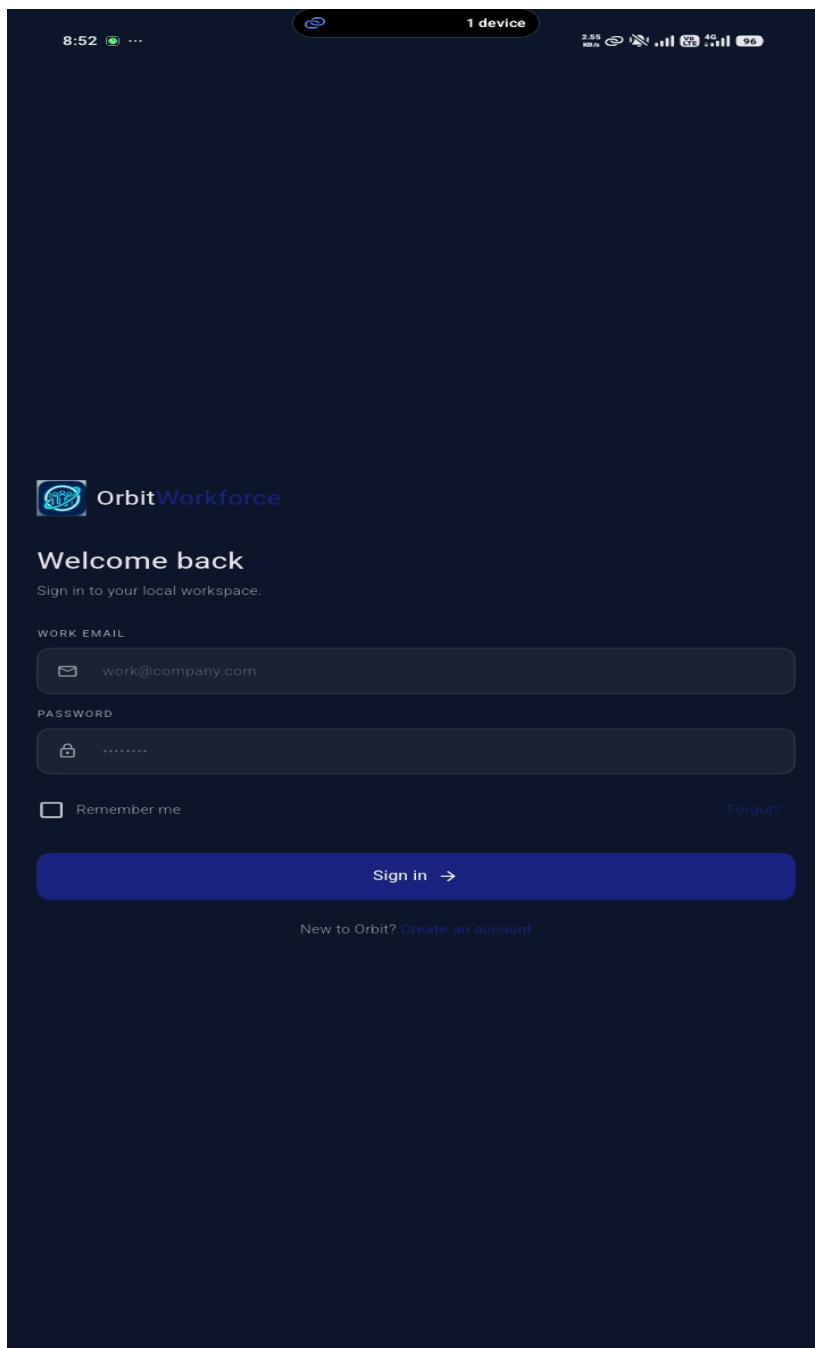


PRESENTATION LAYER

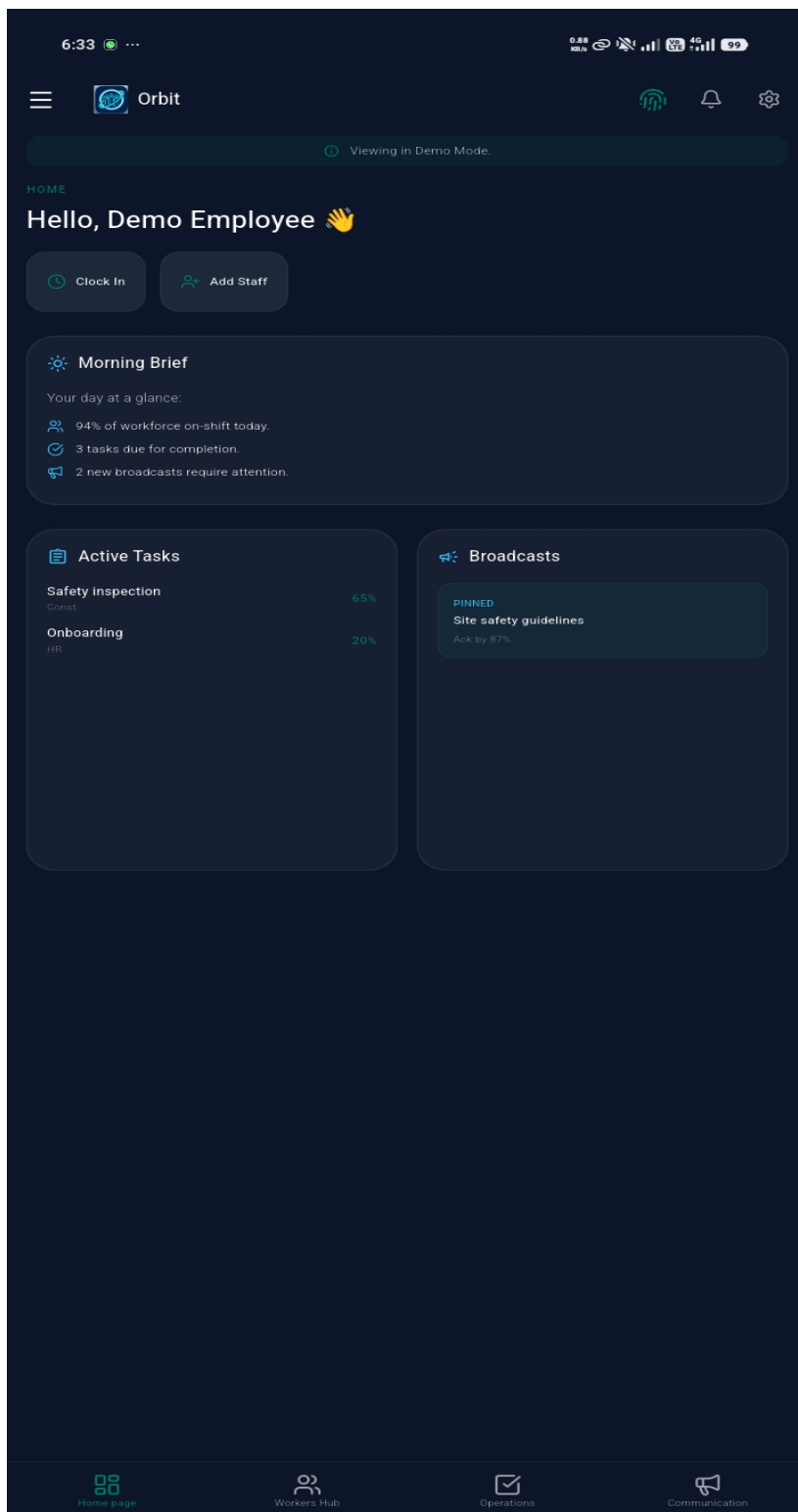
Login Page



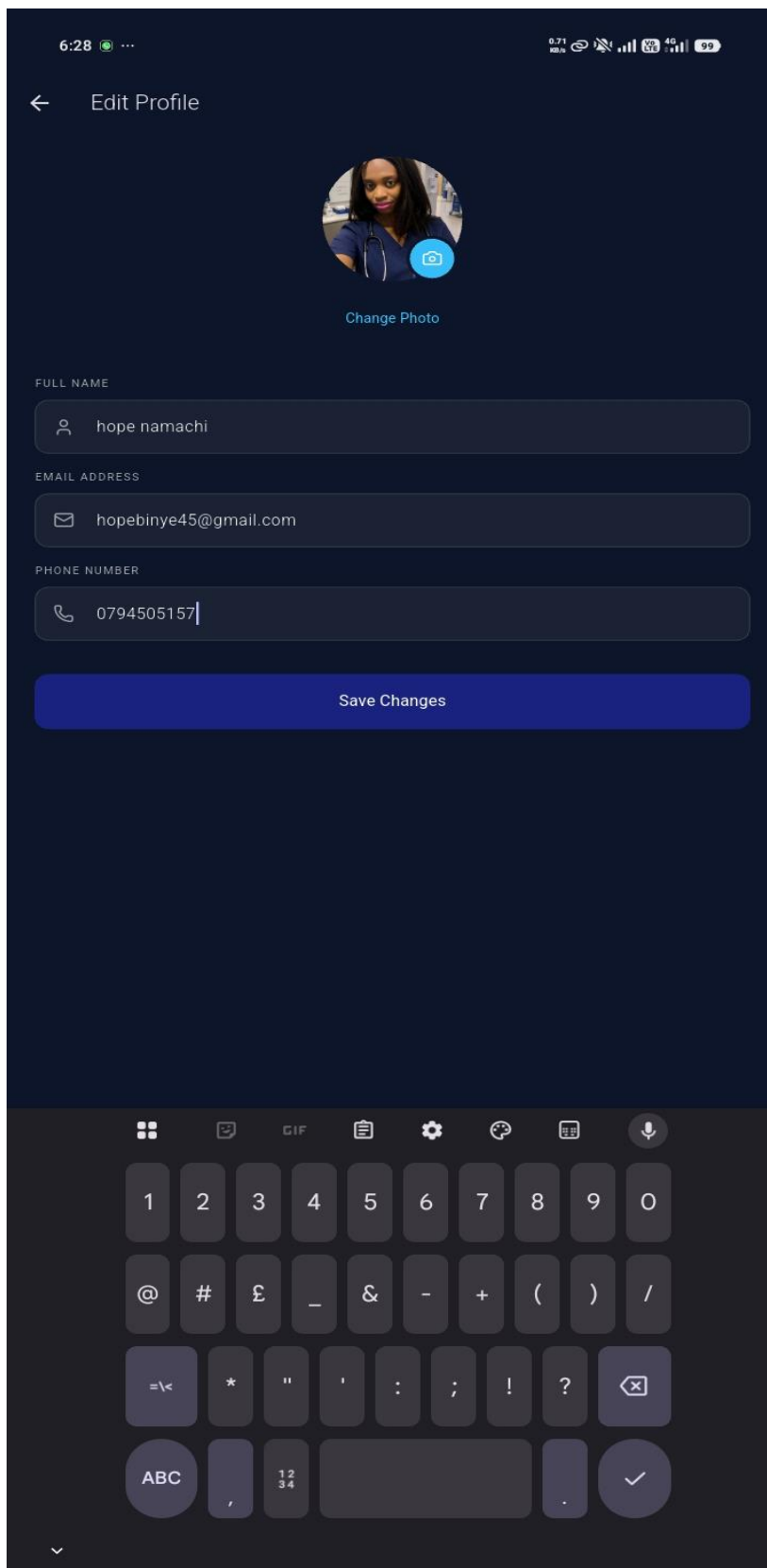
Register page



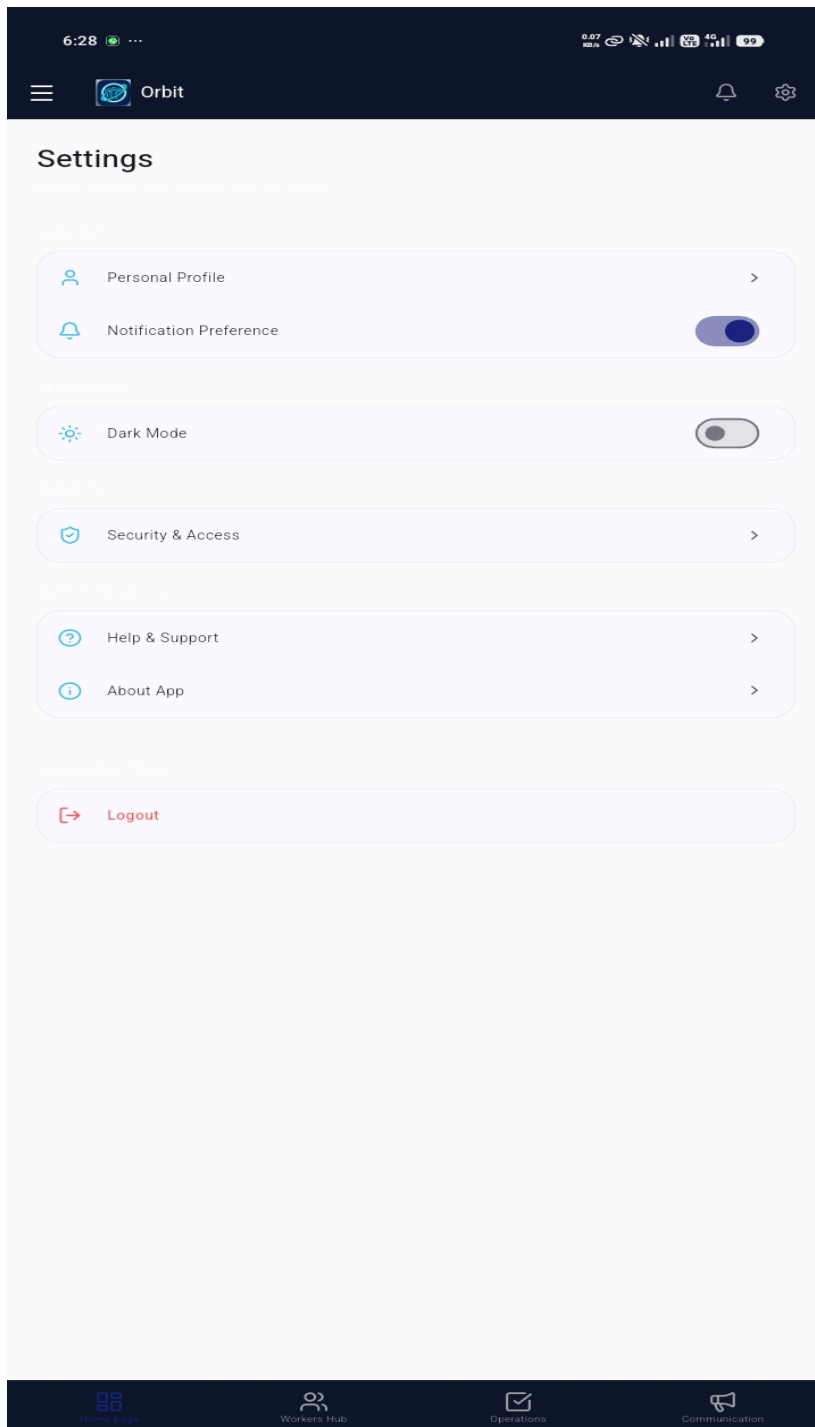
Home page(Dashboard)



Profile Page



Settings Page(in light mode)



BUSINESS LOGIC LAYER

Authentication

Attendance Management

Analytics Processing

Broadcast Management

Workforce Management

↓

DATA LAYER

SQLite Database

Supabase Cloud Database

3.2 Application Framework

Project Structure

```text

lib/

├─ core/

| └─ constants/

| └─ errors/

| └─ utils/

├─ models/

| └─ employee.dart

| └─ attendance.dart

| └─ task.dart

| └─ message.dart

| └─ event.dart

| └─ payroll.dart

├─ services/

| └─ api\_service.dart

| └─ auth\_service.dart

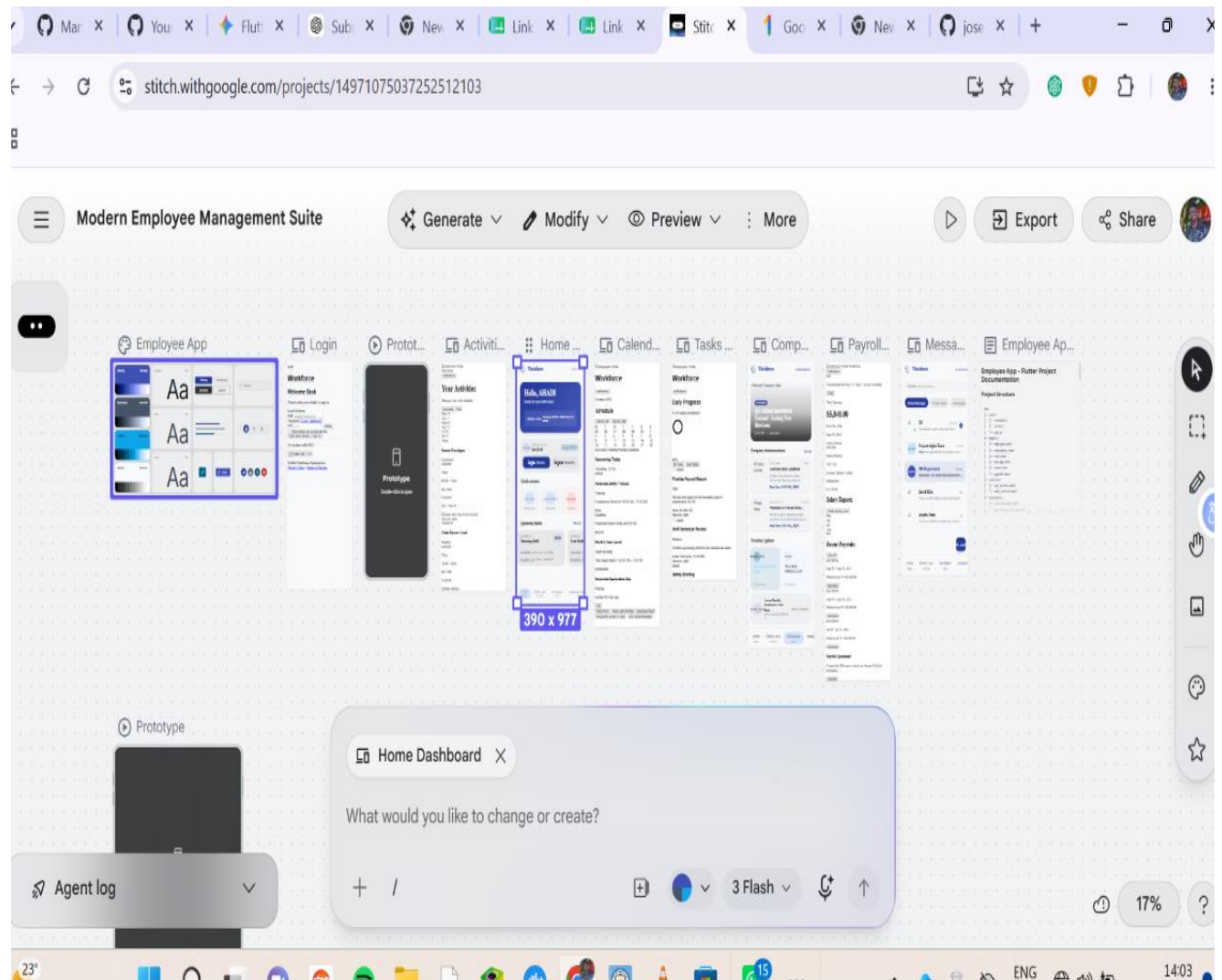
└─ providers/

- | |─ auth\_provider.dart
- | |─ attendance\_provider.dart
- | |─ task\_provider.dart
- | |─ news\_provider.dart
- | |─ chat\_provider.dart
- |─ screens/
  - | |─ auth/
    - | | |─ login\_screen.dart
    - | | |─ signup\_screen.dart
    - | | |─ forgot\_password\_screen.dart
  - | |─ home/
    - | | |─ home\_dashboard.dart
    - | | |─ widgets/
  - | |─ activities/
    - | | |─ activities\_screen.dart
    - | | |─ calendar\_screen.dart
  - | |─ news/
    - | | |─ news\_feed\_screen.dart
  - | |─ tasks/
    - | | |─ task\_list\_screen.dart
  - | |─ payroll/
    - | | |─ payroll\_screen.dart
  - | |─ messages/
    - | | |─ message\_list\_screen.dart
    - | | |─ chat\_detail\_screen.dart
- |─ widgets/
  - | |─ common/



```
| | | └─ custom_button.dart
| | | └─ custom_card.dart
| | | └─ skeleton_loader.dart
| | └─ status_badge.dart
| └─ navigation/
| └─ bottom_nav_bar.dart
| └─ drawer_menu.dart
└─ routes/
 └─ app_router.dart
 └─ theme/
 └─ app_theme.dart
 └─ color_schemes.dart
 └─ main.dart
```

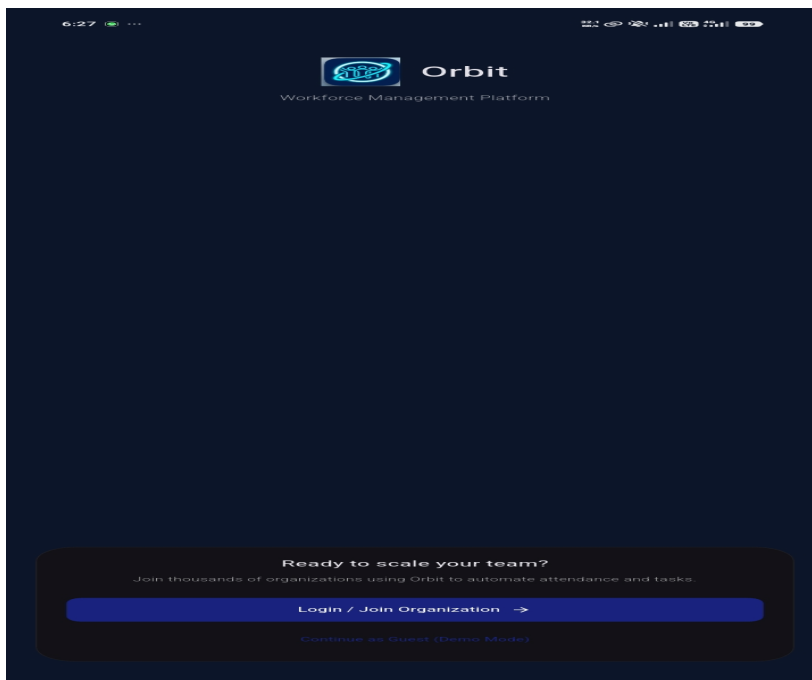
## USED STITCH GOOGLE FRAMEWORK IN PLACE OF FIGMA FRAMEWORK FOR VIRSULIZATION OF HOW SCREEN WERE TO CONNECT BEFORE DEVELOPMENT

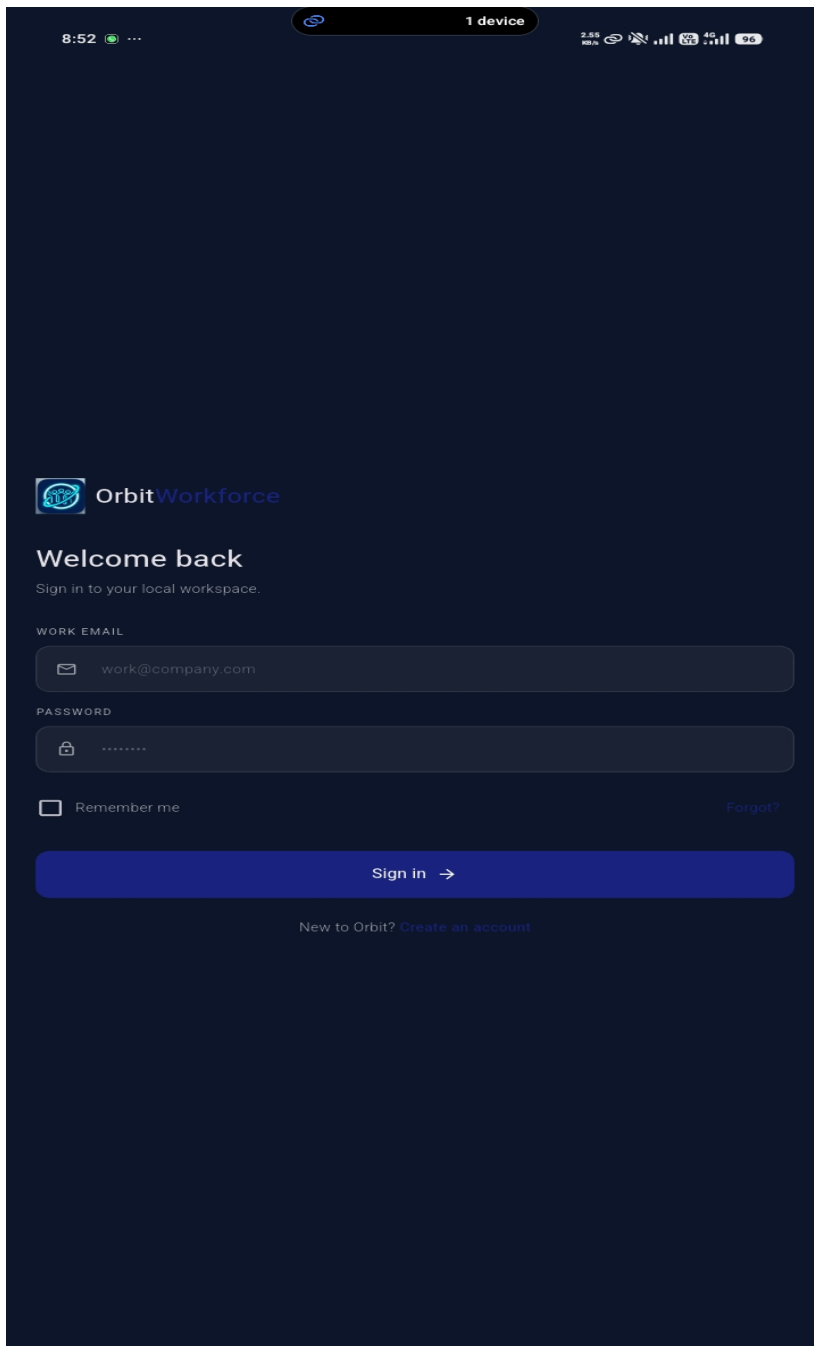


### 3.3 User Interface Design

The application contains the following interfaces: the screens of some are shown above.

## 1. Landing Page





## 2. Login Page

6:27

32.1 MB/s 4G 99



Orbit

Workforce Management Platform

Ready to scale your team?

Join thousands of organizations using Orbit to automate attendance and tasks.

Login / Join Organization →

[Continue as Guest \(Demo Mode\)](#)

3.

8:52

1 device

3.00 MB/s

4G

96

 Orbit

### Create your account

Local SQLite Backend (No WiFi Required)

Create Organization

Join with Code

FULL NAME

 .....

WORK EMAIL

 alex@company.com

PASSWORD

 .....

ORGANIZATION NAME

 Acme Industries

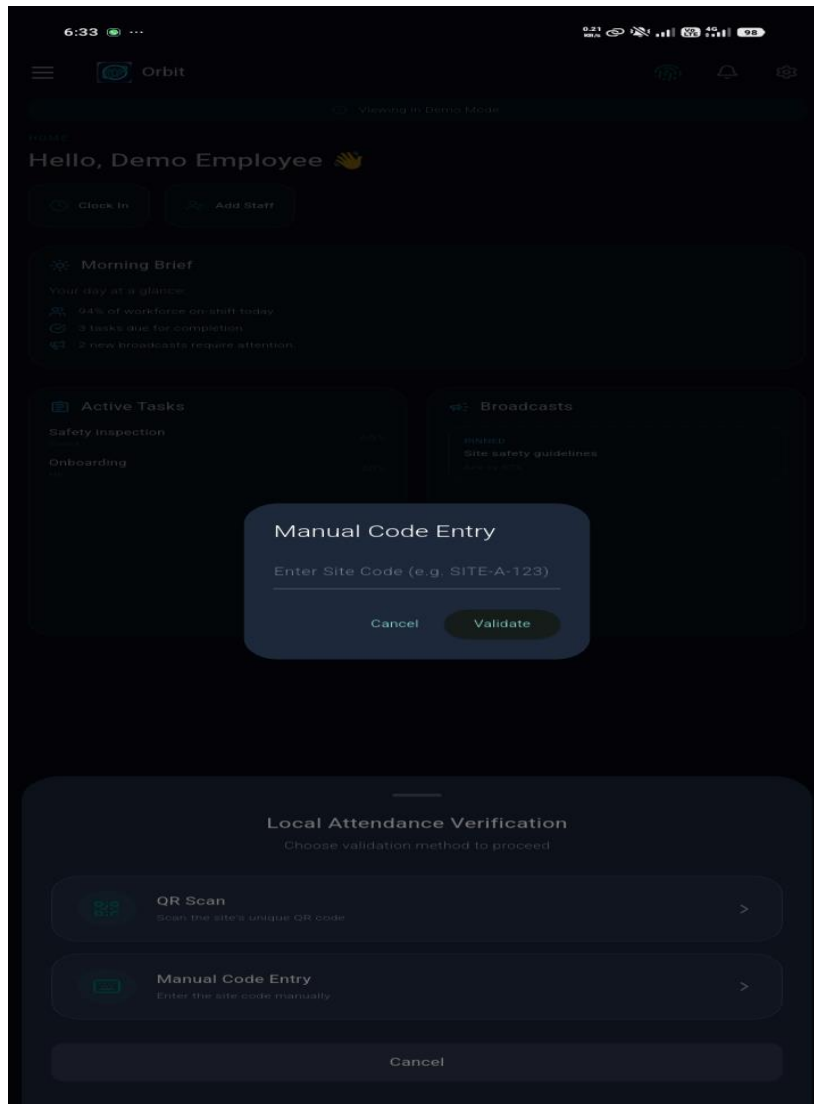
Continue →

Already on Orbit? [Sign in](#)

4.

## Code entry lock in

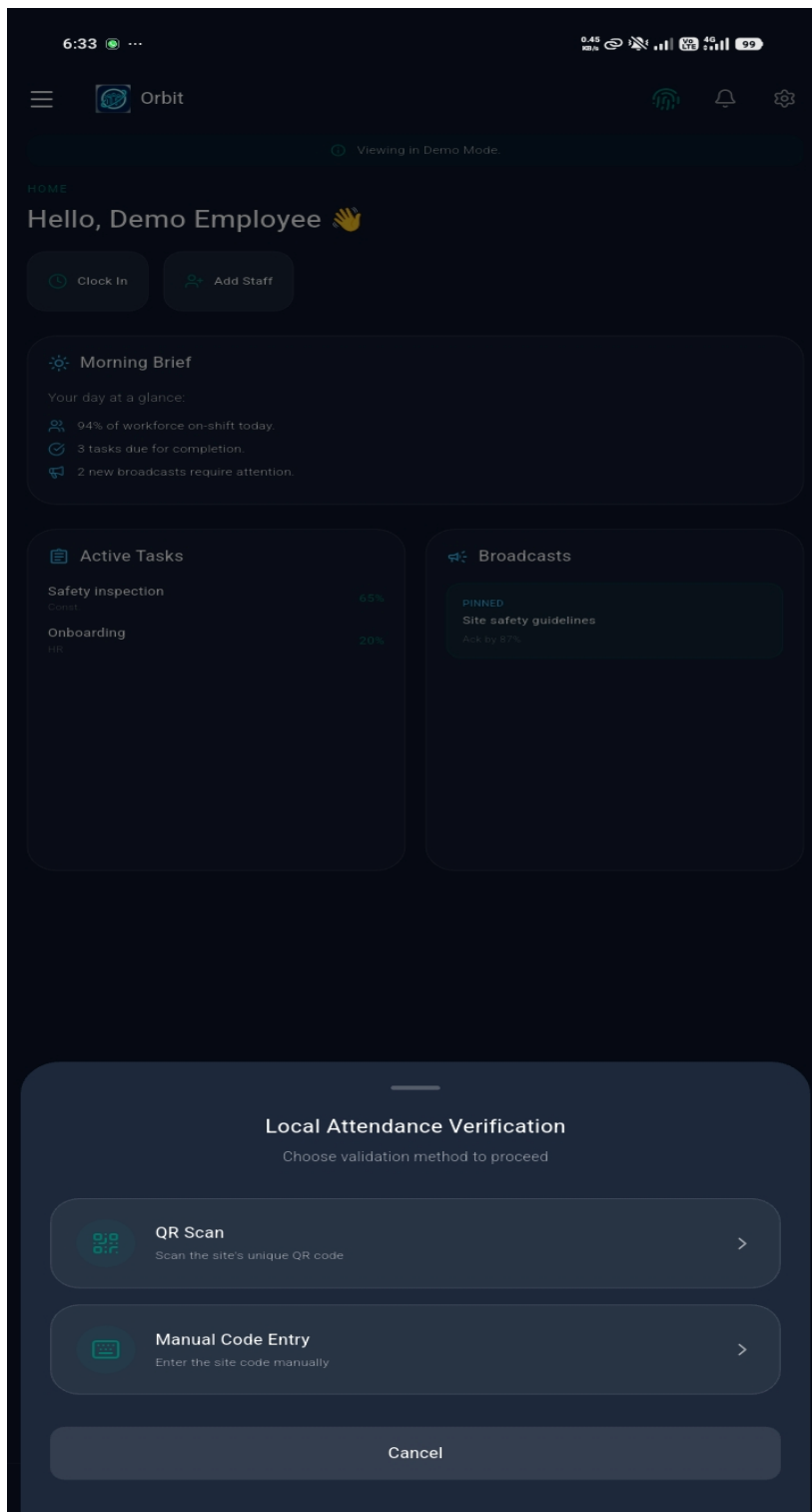
This allows the user to enter the organization for them to be authorized and clock in accepted.



5.

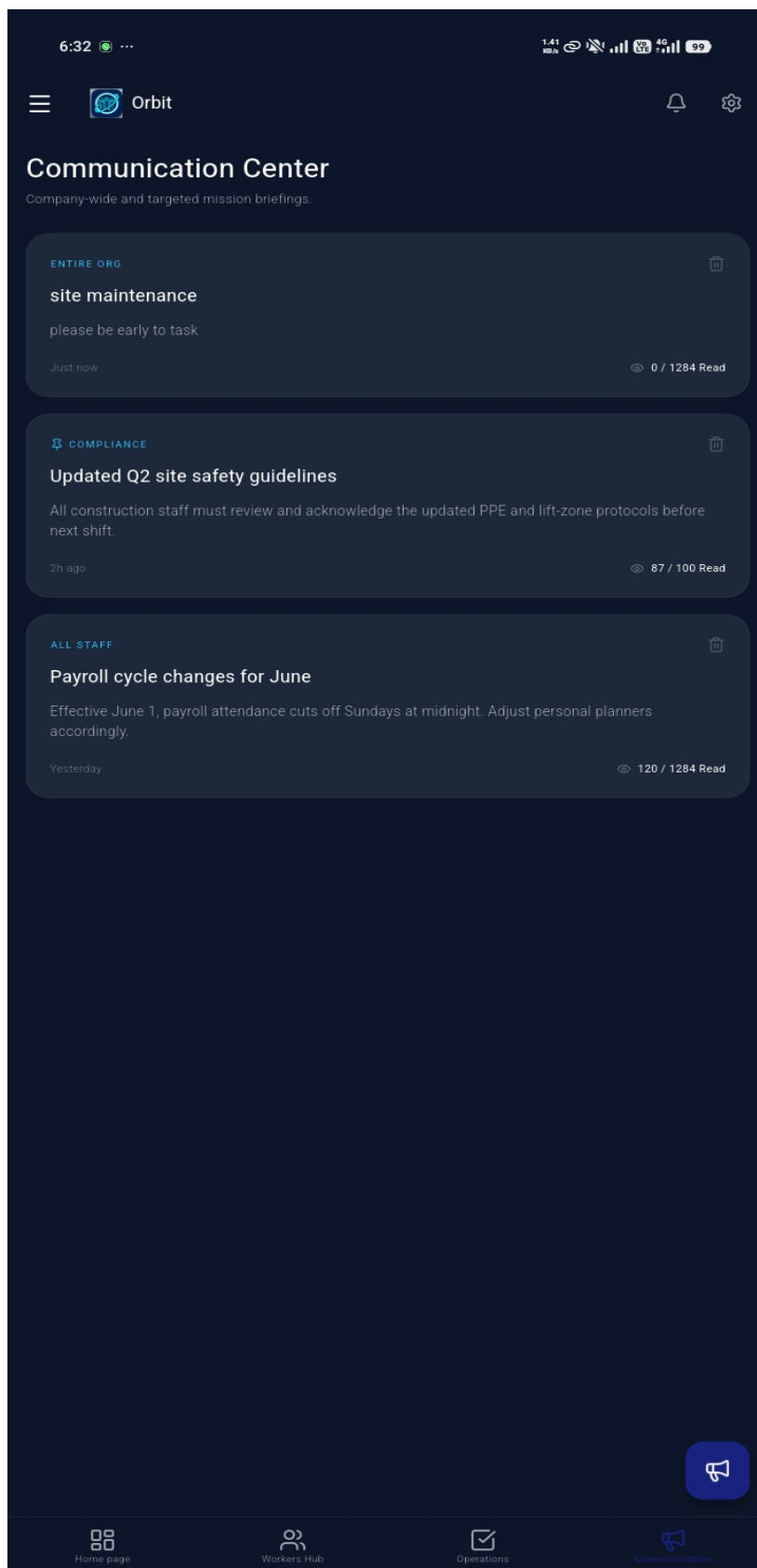
## Other clock in alternatives

After clock in the user can either use a QR scanner that will allow him or her to use a different clock in method



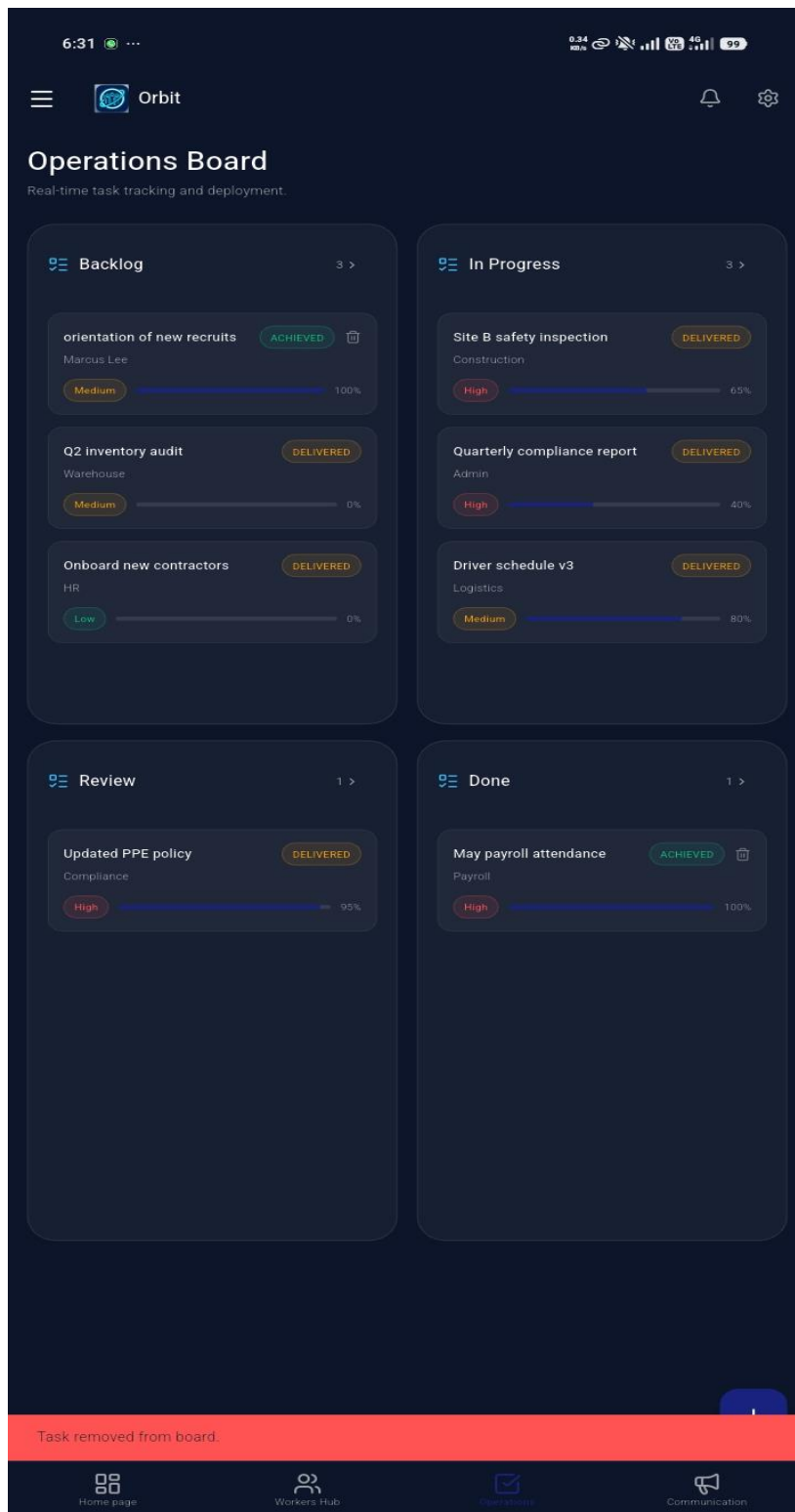


## Communication Centre page



## Operation Board Page

This page allows users employees and managers to upload ongoing operations in the organisations which cannot also be deleted.



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## **Development Tools**

The application was developed using:

- Flutter Framework
- Dart Programming Language
- Android Studio IDE
- SQLite Database
- Supabase Backend Services

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## **Database Management**

The application uses SQLite for local data storage and Supabase for cloud-based services.

Implemented database operations include:

- Create Records
- Read Records
- Update Records
- Delete Records

These CRUD operations allow efficient workforce record management.

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## **Application Modules**

### **Authentication Module**

Responsible for user registration and login.

### **Workforce Module**

Responsible for employee record management.

### **Attendance Module**

Responsible for attendance tracking.

### **Payroll Module**

Responsible for payroll information management.

### **Analytics Module**

Responsible for workforce statistics and reporting.

### **Broadcast Module**

Responsible for organizational communication.

### **Profile Module**

Responsible for user account management.

### **Settings Module**

Responsible for application customization.

### **CHALLENGES ENCOUNTERED**

1. Flutter package dependency conflicts.
2. Database integration challenges.
3. User interface alignment issues.
4. Authentication configuration.
5. Debugging navigation errors.

### **FUTURE IMPROVEMENTS**

1. GPS Attendance Tracking.
2. Push Notifications.
3. Biometric Authentication.
4. Advanced Analytics Dashboard.
5. Cloud Synchronization.
6. AI-Powered Workforce Insights.
7. Multi-Organization Support.